Case study on:
Self-Driving Car Using Obstacle Detection

Subject
Intelligent Systems
Students Name
Bishal kr Thakur(220107)
Santosh Jaiswal (220148)
Program
BEIT / 7th semester
College
Cosmos College of Management and Technology
Date
03/07/2026

Abstract

Autonomous vehicles are one of the most significant applications of intelligent systems. These vehicles use sensors, control algorithms, and decision-making processes to operate without human intervention. The objective of this project is to design and simulate a simple intelligent self-driving car model that can detect obstacles and automatically respond to them.

The system uses an ultrasonic sensor to measure the distance between the vehicle and obstacles. An Arduino microcontroller processes this information and controls the movement of the car using a motor driver and DC motors. When the sensor detects an obstacle within a specific range, the system stops and changes direction automatically.

The simulation of the system is implemented using Tinkercad, which demonstrates the concept of intelligent agents performing perception, decision making, and action.

1. Introduction

Intelligent systems are computer-based systems that can sense their environment, process information, and take actions to achieve specific goals. These systems are widely used in various fields such as robotics, healthcare, transportation, and automation.

One of the most popular applications of intelligent systems is the development of autonomous vehicles. Self-driving cars are designed to navigate roads, detect obstacles, and make driving decisions without human involvement.

Modern autonomous vehicles use technologies such as artificial intelligence, sensors, computer vision, and machine learning to analyze their surroundings. However, the basic concept of an autonomous system can be demonstrated using simple sensors and microcontrollers.

This project focuses on designing a basic intelligent vehicle model that can detect obstacles using an ultrasonic sensor and automatically adjust its movement. The simulation demonstrates how an intelligent system can perceive the environment, process information, and perform actions accordingly.

2. Objectives

The main objectives of this project are:

1. **Understand Intelligent Systems:**
   Learn how sensors, microcontrollers, and actuators work together to make decisions automatically.

2. **Design a Self-Driving Car Model:**
   Build a simple vehicle that can navigate and avoid obstacles using Arduino and DC motors.

3. **Demonstrate Obstacle Detection:**
   Implement ultrasonic sensor-based distance measurement to detect obstacles in real time.

4. **Implement Decision-Making Logic:**
   Program the Arduino to process sensor data and control the motors to move forward, stop, or turn.

5. **Simulate the System in Tinkercad:**
   Use Tinkercad to create a virtual prototype and demonstrate intelligent behavior in a safe, controlled environment.

These objectives aim to provide a practical understanding of intelligent agent behavior and basic autonomous navigation principles.

3. System Components

The following hardware components are used in the simulation:

Component	Description
Arduino UNO	Microcontroller used for processing sensor data
Ultrasonic Sensor (HC-SR04)	Detects obstacles and measures distance
L293D Motor Driver	Controls the direction of DC motors
DC Motors	Used to move the vehicle
Breadboard	Used for circuit connections
Power Supply / Battery	Provides power to the system

Each component plays an important role in enabling the system to function as an autonomous vehicle.

4. System Architecture

The self-driving car system is designed using a modular approach, consisting of three main components:

1. Sensor Module

- Uses an ultrasonic sensor (HC-SR04) to detect obstacles in front of the vehicle.
- Continuously measures the distance between the vehicle and nearby objects.
- Sends the measured distance data to the Arduino microcontroller for processing.

2. Processing Module

- The Arduino UNO acts as the intelligent agent.
- Receives sensor data and makes decisions based on programmed logic.
- Determines whether the vehicle should move forward, stop, or change direction.
- Implements real-time decision-making, demonstrating the perception-decision-action cycle of intelligent systems.

3. Actuator Module

- Consists of the L293D motor driver and DC motors.
- Executes the movement commands from the Arduino:
    - Move Forward: Both motors rotate in forward direction.
    - Stop: Motors are turned off.
    - Turn/Change Direction: Motors rotate in opposite directions to turn the vehicle.
- Converts the microcontroller's decisions into physical actions.

5. Block Diagram

The block diagram of the system is shown below.

```
Environment
      ↓
Ultrasonic Sensor
      ↓
Arduino Microcontroller
(Decision Making Unit)
      ↓
Motor Driver (L293D)
      ↓
DC Motors
      ↓
Vehicle Movement
```

- The block diagram represents the overall structure of the self-driving car system.
- The **ultrasonic sensor** detects obstacles and measures the distance from objects.
- The sensor sends the distance data to the **Arduino UNO**.
- Arduino processes the data and decides the movement of the car.
- The decision signals are sent to the **L293D motor driver**.
- The motor driver controls the **DC motors** to move or stop the vehicle.

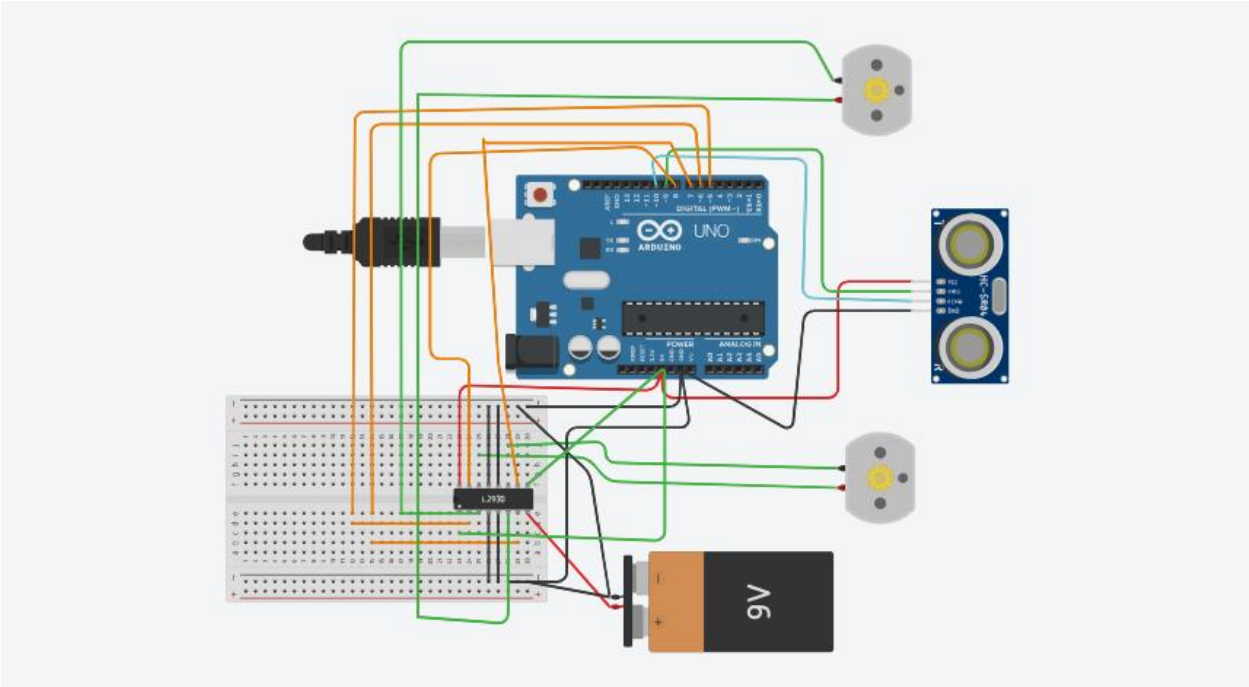
6. Flowchart of System Operation

The flowchart below shows the working logic of the autonomous system.

```
Start
  ↓
Initialize Sensor and Motors
  ↓
Measure Distance from Obstacle
  ↓
Is Distance < 20 cm?
  ↓              ↓
Yes              No
 ↓                ↓
Stop Car      Move Forward
 ↓
Turn Direction
 ↓
Repeat Process
```

- The system starts and initializes the Arduino, sensor, and motor driver.
- The **ultrasonic sensor** measures the distance of obstacles continuously.
- The measured distance is sent to the **Arduino** for processing.
- If the distance is greater than the threshold value, the car moves forward.
- If the distance is less than the threshold value, the car stops or changes direction.
- The process repeats continuously to avoid obstacles.

7. Circuit Diagram



The circuit diagram shows the connections between Arduino UNO, ultrasonic sensor, motor driver, and DC motors.

Main connections include:

- Ultrasonic sensor connected to Arduino pins for distance measurement.
- Motor driver connected to Arduino to control motor direction.
- DC motors connected to the output pins of the motor driver.

The circuit enables the microcontroller to control the movement of the motors based on sensor input.

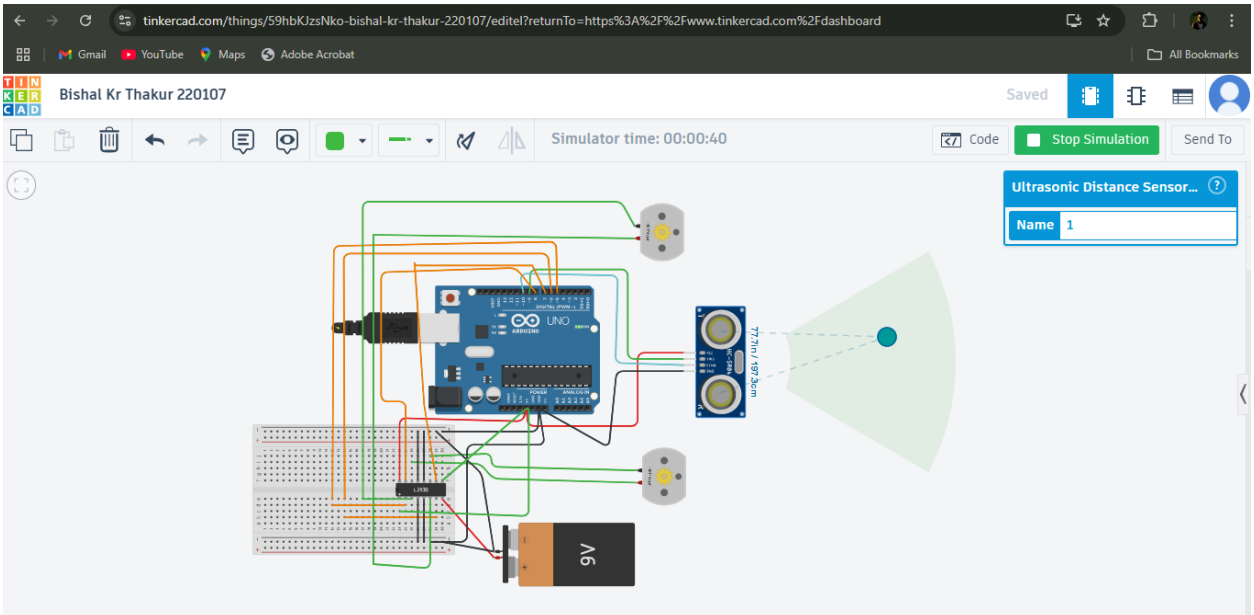
8. Arduino Program Code

```
#define trigPin 9
#define echoPin 10

#define IN1 5
#define IN2 6
#define IN3 7
#define IN4 8

long duration;
int distance;

void setup() {

  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);

  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);
  pinMode(IN3, OUTPUT);
  pinMode(IN4, OUTPUT);

  Serial.begin(9600);
}

void loop() {

  distance = getDistance();

  Serial.print("Distance: ");
  Serial.println(distance);

  if (distance > 20) {
    moveForward();
  }
  else {
    stopCar();
```

```
    delay(500);

    turnRight();
    delay(700);
  }

  delay(100);
}


int getDistance() {

  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);

  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  duration = pulseIn(echoPin, HIGH);

  distance = duration * 0.034 / 2;

  return distance;
}


void moveForward() {

  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);

  digitalWrite(IN3, HIGH);
  digitalWrite(IN4, LOW);
}
```

```
void stopCar() {
digitalWrite(IN1, LOW);
  digitalWrite(IN2, LOW);

  digitalWrite(IN3, LOW);
  digitalWrite(IN4, LOW);
}
void turnRight() {
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, HIGH);

  digitalWrite(IN3, HIGH);
  digitalWrite(IN4, LOW);
}
```

9. Simulation Result



The simulation demonstrates the working of the obstacle-avoiding vehicle.

Observed behavior:

- The vehicle moves forward when no obstacle is detected.
- When an obstacle appears within 20 cm, the vehicle stops.
- The system then changes direction to avoid collision.
  This behavior illustrates how an intelligent system can automatically react to environmental conditions.

10. Applications

The proposed self-driving car model demonstrates how an intelligent agent can sense its environment, make decisions, and act automatically. This concept is widely applicable in real life, including:

- **Autonomous Vehicles:** Shows the basic principle behind self-driving cars that can avoid obstacles without human intervention.
- **Robotics:** Useful for mobile robots, delivery robots, or warehouse robots that need to navigate safely.
- **Obstacle Avoidance Systems:** The sensor-based approach is the foundation for smart toys, robotic vacuum cleaners, and autonomous drones.
- **Educational Purpose:** Helps students understand intelligent systems, agent-environment models, and real-time decision-making.
- **Smart Transportation:** Can be extended to traffic management, autonomous delivery systems, and other smart mobility solutions.

This application section highlights how a simple simulation can demonstrate the fundamentals of intelligent systems and their potential real-world impact.

.

11. Limitations

Although this self-driving car project successfully demonstrates the basic principles of intelligent systems, it has some limitations:

1. **Limited Sensor Range:**
   The ultrasonic sensor can only detect obstacles within a certain distance (approximately 2–400 cm).
2. **No Complex Obstacle Handling:**
   The system cannot handle multiple obstacles simultaneously or complex environments with curves and intersections.
3. **No Advanced AI or Learning:**
   The Arduino follows pre-programmed logic and cannot learn from experience or adapt to new situations.
4. **Simulation Only:**
   The project is implemented and tested in Tinkercad simulation; it may behave differently in a real-world environment.
5. **Basic Navigation:**
   The car can only move forward, stop, and turn right; it does not include advanced maneuvers like turning left or reversing intelligently.
   Despite these limitations, the project effectively demonstrates the fundamental concepts of an intelligent obstacle-avoiding agent.

12. Conclusion

This project successfully demonstrates the design and simulation of a simple intelligent self-driving car system. The system uses an ultrasonic sensor to detect obstacles and an Arduino microcontroller to process the information and control the motors.

The simulation shows how an intelligent agent can perceive the environment, make decisions, and perform actions automatically. Although the model is simple, it effectively illustrates the fundamental principles of intelligent systems and autonomous vehicle technology.

Future improvements may include adding multiple sensors, camera-based vision systems, and machine learning algorithms to enhance the system's capabilities.

References

1. Simon, H. A. (1996). *The Sciences of the Artificial*. MIT Press.
2. Raj, P., & Krishnan, S. (2020). *Arduino Projects for Beginners: Intelligent Robots and Automation*. Packt Publishing.
3. Tinkercad Circuits. Autodesk. Available at: https://www.tinkercad.com
4. HC-SR04 Ultrasonic Sensor Datasheet. SparkFun Electronics. Available at: https://www.sparkfun.com/datasheets/Sensors/Proximity/HCSR04.pdf
5. L293D Motor Driver Datasheet. Texas Instruments. Available at: https://www.ti.com/lit/ds/symlink/l293.pdf
6. Arduino Documentation. Arduino Official Website. Available at: https://www.arduino.cc/en/Guide/HomePage

These references cover the **theoretical background of intelligent systems, microcontroller programming, sensor usage, and motor control**, all of which were used in this project.